Nishka Mathew

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SUMMARY

CS + Psychology graduate from UNC Charlotte. Interested in UX Research, Machine Learning, and Data Analyst Roles.

EDUCATION

The University of North Carolina at Charlotte ***GPA: 3.83***

Bachelor of Arts in Computer Science and Bachelor of Science in Psychology

- **Relevant Coursework**: Research Methodology II, Data Structures & Algorithms, Data Mining, Human-Centered AI, Cognitive Science, iOS Development, Design Prototyping, UX Methods, Elements of Statistics II

Honors: Chancellor's List (Fall 2022 - Fall 2023, Fall 2024, Fall 2025) & Dean's List (Spring 2024, Spring 2025)

WORK EXPERIENCE

Shift Lead, Server & Bartender at Jinya Ramen Bar | *Charlotte, NC* **September 2024 – Current**

- Led front-of-house operations for a **15**-member team during high-volume service to ensure efficient workflow
- Onboarded and mentored new team members on service and operational procedures to the highest standards
- Made data-informed decisions by tracking table turnover, inventory, and peak service to optimize resource allocation

Research Intern at Security Privacy and User Experience (SPEX) Lab | *Charlotte, NC* **May 2024 – July 2025**

- Coauthored a research paper published to **ACM COMPASS 2025** titled *Privacy on the Move: Understanding Educational Migrants' Social Media Practices through the Lens of Communication Privacy Management Theory*
- Conducted **25+** semi-structured interviews with first-generation international students from over **14** countries
- Analyzed qualitative data using inductive coding to identify **5** key themes shaping students' privacy practices
- Utilized **IBM SPSS Statistics** to analyze quantitative data and created a **Top-Line Report** with **10+** visualizations

Teacher's Assistant at the UNC Charlotte | *Charlotte, NC* **January 2023 – May 2024**

Spring 2023: College Algebra, **Fall 2023**: Research Methodology I**, Spring 2024**: Research Methodology II

- Facilitated workshops and tutoring sessions for **200+** students across **3** courses, improving performance by **12-15%**
- Provided feedback on **30+** student research projects through every stage, guiding proposals through a mock IRB submission process to ensure ethical compliance

PROJECTS

Pearl Hacks 2025 – 1st Place Sponsored by Qorvo

- Built a responsive **HTML/CSS/JavaScript** platform that connects industry mentors, researchers, and students
- Features include mentorship matching with filters, a research collaboration portal, and daily career-focused Q&A

AI-Powered Microalgae Biomass Predictor for Ecoclosure Biochromic Windows

- Developed predictive machine learning models to forecast microalgae biomass growth in Ecoclosure's Biochromic Window systems, leveraging multivariate environmental sensor data to optimize energy and carbon capture
- Tested **linear, polynomial, logarithmic,** and **random forest regression models** on a Kaggle sample dataset (9,784 data points) and real-world Innovation Barn sensor data in **Jupyter Notebook** using **Python, Scikit-learn**, and **Pandas**
- Designed an interactive visualization framework overlaying historical biomass trends with **14-day predictive projections**, enabling non-technical stakeholders to interpret model outputs and understand multivariate relationships
- Contributed to a human-centered AI pipeline combining explainable ML with time-series data visualization

CinemaQuest – Movie Discovery & Watchlist iOS App

- Designed and developed an **iOS** app using **Xcode** and **Swift** that allows users to discover trending movies, manage personal watchlists, and track watched titles, integrating The Movie Database (TMDB) **API** for up-to-date movie data
- Built an intuitive UI with **UIKit** components, asynchronous image loading (**Nuke**), and user data persistence

SKILLS

Python, SQL, Pandas, Scikit-learn, Jupyter Notebook, Machine Learning, IBM SPSS Statistics, Qualtrics, Overleaf, CITI Human Subjects Research Certification, Postman, Tableau, PowerBi, HTML, Figma, Javascript